

Street Light Monitoring System



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01

Hardware & Maquette

- Controller: ESP32-D0WD-V3
- Ambient Sensing: 1 × analog LDR (GPIO 34) → day/night via 12-bit ADC
- Lamp Simulation: 3 × PWM LEDs (GPIO 2, 5, 4), brightness inverse to ambient light
- Fault Detection: 3 × digital LDRs (GPIO 27, 14, 13) → HIGH when LED “off” → flags broken
- Prototype: All on a compact breadboard/maquette for demo.



02

MQTT & Broker

- Protocol: MQTT—ultra-lightweight pub/sub for IoT.
- Broker: Mosquitto v2.0.21 on laptop (ports 1883 TCP, 9001 WS).
- Port Forwarding: Router forwards public TCP 1883 to laptop’s LAN IP.
- WAN Access: Any MQTT client uses your public IP + 1883 to publish/subscribe.

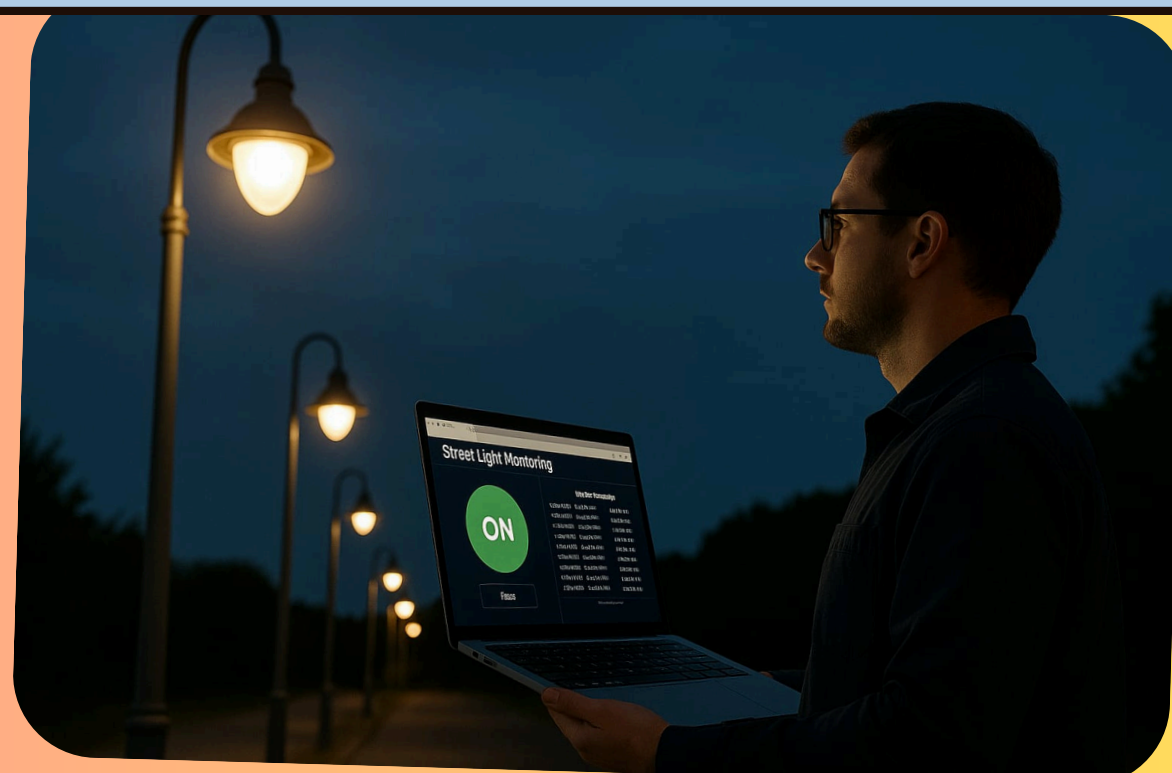
MQTT

Web Server & Dashboard

- Backend: Python Flask + paho-mqtt, secured with HTTP Basic Auth
- Subscription: listens on esp32/bruh/status, updates in-memory lamp_status & log (last 10 msgs)
- UI: dynamic HTML with JavaScript auto-refresh (5 s), status cards (blue=OK, red=broken) and per-lamp reset buttons
- Hosting: serves on port 5000; WAN access via router port-forward (public_IP:5000)

03

The system uses an ESP32-D0WD-V3 to monitor ambient light via an analog LDR and control three PWM-based LED street-lamp simulators. It checks lamp functionality using three digital LDRs. Brightness data and status updates are sent to an Eclipse Mosquitto MQTT broker on a laptop, which is accessible online. A secure Flask web dashboard (public IP, port 5000) displays real-time lamp status, logs the last ten messages, and includes lamp reset controls.



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